

Page 1: Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include reasoning, learning, problem-solving, perception, and language understanding. AI is used in search engines, recommendation systems, autonomous vehicles, healthcare, finance, and education. Artificial Intelligence (AI) is the field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include reasoning, learning, problem-solving, perception, and language understanding. AI is used in search engines, recommendation systems, autonomous vehicles, healthcare, finance, and education. Artificial Intelligence (AI) is the field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include reasoning, learning, problem-solving, perception, and language understanding. AI is used in search engines, recommendation systems, autonomous vehicles, healthcare, finance, and education.

Page 2: Machine Learning

Machine Learning (ML) is a subset of AI where computers learn patterns from data instead of being explicitly programmed for every rule. Common types include supervised, unsupervised, and reinforcement learning. Popular libraries include scikit-learn, TensorFlow, and PyTorch. Machine Learning (ML) is a subset of AI where computers learn patterns from data instead of being explicitly programmed for every rule. Common types include supervised, unsupervised, and reinforcement learning. Popular libraries include scikit-learn, TensorFlow, and PyTorch. Machine Learning (ML) is a subset of AI where computers learn patterns from data instead of being explicitly programmed for every rule. Common types include supervised, unsupervised, and reinforcement learning. Popular libraries include scikit-learn, TensorFlow, and PyTorch.

Page 3: Generative AI and LLMs

Generative AI creates new content such as text, images, code, audio, and video. Large Language Models (LLMs) are trained on vast amounts of text and can answer questions, summarize documents, write code, and assist with research. Retrieval-Augmented Generation (RAG) combines LLMs with external knowledge sources to improve factual accuracy. Generative AI creates new content such as text, images, code, audio, and video. Large Language Models (LLMs) are trained on vast amounts of text and can answer questions, summarize documents, write code, and assist with research. Retrieval-Augmented Generation (RAG) combines LLMs with external knowledge sources to improve factual accuracy. Generative AI creates new content such as text, images, code, audio, and video. Large Language Models (LLMs) are trained on vast amounts of text and can answer questions, summarize documents, write code, and assist with research. Retrieval-Augmented Generation (RAG) combines LLMs with external knowledge sources to improve factual accuracy.

Page 4: AI Applications

AI is transforming healthcare through medical image analysis and drug discovery, finance through fraud detection, education through personalized learning platforms, manufacturing through predictive maintenance, and customer service through chatbots and virtual assistants. AI is transforming healthcare through medical image analysis and drug discovery, finance through fraud detection, education through personalized learning platforms, manufacturing through predictive maintenance, and customer service through chatbots and virtual assistants. AI is transforming healthcare through medical image analysis and drug discovery, finance through fraud detection, education through personalized learning platforms, manufacturing through predictive maintenance, and customer service through chatbots and virtual assistants.

Page 5: Future and Ethics

As AI becomes more capable, responsible development is increasingly important. Key topics include privacy, fairness, transparency, bias mitigation, security, and human oversight. The future of AI is expected to include more capable multimodal systems, intelligent agents, and broader integration into everyday software. As AI becomes more capable, responsible development is increasingly important. Key topics include privacy, fairness, transparency, bias mitigation, security, and human oversight. The future of AI is expected to include more capable multimodal systems, intelligent agents, and broader integration into everyday software. As AI becomes more capable, responsible development is increasingly important. Key topics include privacy, fairness, transparency, bias mitigation, security, and human oversight. The future of AI is expected to include more capable multimodal systems, intelligent agents, and broader integration into everyday software.